

IT 520 — Friday Practice Quiz

Friday, September 18, 2026 • Revised practice for Monday, September 21

30 multiple-choice questions • Select one answer, A–E, for each question.

Target completion time: 30 minutes. You may use the full 50-minute period.

Use scratch paper for calculations. All numerical scenarios are hypothetical. Keep units consistent. Questions within a named scenario share the facts in its scenario box; each answer can still be found without using an earlier answer.

Name: _____ Date: _____

Scenario 1 — Follow the payroll job

A payroll export begins as a stored file. The program loads records into RAM, the CPU processes them, the result remains in RAM, and the finished report is saved. Before one instruction runs: PC = 40, memory[40] = ADD 1 TO [201], and memory[201] = 7.

1. Which sequence best follows the job through the simplified machine?

- A. Storage → CPU → output, skipping working memory.
- B. RAM → storage → CPU, starting after the load.
- C. Storage → RAM (load) → CPU/memory → storage/output (save).
- D. Output → CPU/memory → storage, reversing the path.
- E. Storage → output → RAM → CPU, bypassing the load.

2. Immediately after fetch completes, which state is expected?

- A. IR contains ADD 1 TO [201], PC is 41, and memory[201] is still 7.
- B. IR contains 8, PC is 40, and memory[201] is still 7.
- C. IR is empty, PC is 41, and memory[201] has become 8.
- D. IR contains ADD 1 TO [201], PC is 40, and memory[201] has become 8.
- E. IR contains 7, PC is 201, and memory[40] has become 8.

3. Which description correctly completes decode, execute, and write-back for this instruction?

- A. The decode step stores 8; the ALU advances the PC; write-back reads address 40.
- B. The decode step advances the PC; the ALU stores 7; write-back calculates 8.
- C. The decode step calculates 8; the ALU fetches address 40; write-back clears the IR.
- D. The decode step changes address 201; the ALU reads the PC; write-back identifies ADD.
- E. The decode step decodes ADD and [201]; the ALU makes 8; write-back stores 8.

4. The whole export takes 24 seconds. What measurement would best test whether loading the input file consumes most of that time?

- A. Read the processor's maximum GHz rating and infer the load time.
- B. Place timestamps immediately before and after the load, then repeat comparable runs.
- C. Measure only the saved report's file size after one run.
- D. Count the instructions in one function and call that export latency.
- E. Use the 24-second total as the load time without another observation.

5. Why can the same memory bit pattern serve as an instruction in one moment and data in another?

- A. Instruction patterns usually contain more bits than data patterns.
- B. The ALU permanently labels a pattern when it enters RAM.
- C. Data patterns belong on persistent storage rather than in RAM.
- D. Context determines whether the pattern is used as instruction or data.
- E. The pattern changes automatically when the PC points to it.

Short checks — Machine concepts and evidence

6. Why is a CPU–memory–storage–I/O diagram useful even though it omits many details?

- A. It preserves relevant parts while hiding unnecessary detail.
- B. It includes the electrical events that occur inside a computer.
- C. It makes measurement boundaries unnecessary for performance claims.
- D. It shows the four pictured parts contribute equal amounts of time.
- E. It remains sufficient for unrelated questions about other computers.

7. Unsaved edits exist only in RAM when power is lost. Why might the saved file not include them?

- A. Persistent storage loses its contents whenever power is removed.
- B. The program counter keeps a permanent copy of each edit.
- C. RAM is volatile, and the edits had not reached persistent storage.
- D. The ALU deletes results after completing an addition.
- E. Input/output devices cannot communicate with persistent storage.

8. Which assignment matches the class's simplified CPU model?

- A. PC: current instruction; IR: arithmetic result; ALU: next address.
- B. PC: next address; IR: current instruction; ALU: arithmetic.
- C. PC: saved file; IR: display pixels; ALU: RAM capacity.
- D. PC: arithmetic result; IR: next address; ALU: current instruction.
- E. PC: transfer rate; IR: storage address; ALU: clock ceiling.

9. During one machine cycle the PC changes from 40 to 41. In the class's model, what performs that update?

- A. A separate instruction stored at address 41 whose job is to update the PC.
- B. The IR, which takes over holding the next address after decode.
- C. Write-back, which stores the value 41 into memory address 201.
- D. An operating-system routine that assigns a next address to each program.
- E. The fetch step itself, which advances the PC as part of the cycle.

10. A buyer says a chip with twice as many transistors must finish the payroll export in half the time. What evidence would actually test that claim?

- A. The newer chip's launch date and purchase price.
- B. The number of cores printed on either package.
- C. A statement that Moore's law applies uniformly across programs.
- D. Timed runs of the same export under comparable conditions.
- E. A larger transistor count with no run of the export.

Scenario 2 — Upgrade one stage of a report

A report has a 6-second CPU stage followed by an 18-second transfer stage; the stages do not overlap. The CPU stage runs 3 billion instructions at an average of 4 cycles per instruction: 12 billion cycles total. It currently sustains 2 GHz. A proposed CPU sustains 4 GHz on this stage, with the same instruction and cycle counts.

11. At 2 GHz, how long is one clock cycle?

- A. 2 nanoseconds.
- B. 5 nanoseconds.
- C. 0.2 nanoseconds.
- D. 0.5 nanoseconds.
- E. 2 seconds.

12. Using $\text{cycles} \div \text{cycles per second}$, how long does the CPU stage take?

- A. 24 seconds.
- B. 6 seconds.
- C. 14 seconds.
- D. 0.167 seconds.
- E. 10 seconds.

13. If only the CPU stage changes, what is the report's new total time?

- A. 12 seconds.
- B. 18 seconds.
- C. 9 seconds.
- D. 15 seconds.
- E. 21 seconds.

14. What is the whole-job speedup, using $\text{old total time} \div \text{new total time}$?

- A. About 1.14×
- B. Exactly 2×
- C. About 0.88×
- D. Exactly 3×
- E. About 1.50×

15. Which assumption makes the predicted 3-second CPU stage defensible?

- A. The report also receives twice as much RAM capacity.
- B. The 4 GHz value is a maximum printed on the box.
- C. The cycle count stays fixed and 4 GHz is sustained.
- D. The transfer stage becomes faster when CPU rate rises.
- E. An instruction requires one clock cycle for this workload.

Scenario 3 — Read one service log

During a stable 60-minute observation, a one-worker repair service completes 12 tickets. The worker is busy for 36 minutes. A ticket spends an average of 10 minutes inside the boundary from arrival through completion. Use X = completions/time, $U = X \cdot S$, and $N = X \cdot R$.

16. What is the observed throughput X ?

- A. 12 tickets/hour.
- B. 36 tickets/hour.
- C. 0.2 tickets/hour.
- D. 5 tickets/hour.
- E. 60 tickets/hour.

17. What is the worker's utilization during the observation?

- A. 36%.
- B. 24%.
- C. Exactly 60%.
- D. 167%.
- E. 72%.

18. What is the average service time S per completed ticket?

- A. 12 minutes/ticket.
- B. 5 minutes/ticket.
- C. 36 minutes/ticket.
- D. 0.33 minutes/ticket.
- E. 3 minutes/ticket.

19. Using Little's Law, what is the average number N of tickets inside the boundary?

- A. 120 tickets.
- B. 2 tickets.
- C. 10 tickets.
- D. 1.2 tickets.
- E. 6 tickets.

20. A student computes $60/12 = 5$ minutes and calls that every ticket's latency. What does 5 minutes actually describe here?

- A. The worker's idle time during each ticket.
- B. The number waiting at a typical instant.
- C. The service time of each completed ticket.
- D. Average completion interval, not response time.
- E. The observation duration after removing busy time.

Scenario 4 — Diagnose and improve a report service

During a 60-minute stable observation, a report service completes 12 reports. Busy times are CPU 24 minutes, disk 48 minutes, and network 36 minutes. There is one server at each resource. Service demand is busy time $\div$ completions; a resource's ceiling is $60 \div$ demand.

21. What are the CPU, disk, and network demands per report?

- A. CPU 24 min/report; disk 48; network 36.
- B. CPU 2 min/report; disk 4; network 3.
- C. CPU 40 min/report; disk 80; network 60.
- D. CPU 5 min/report; disk 2.5; network 3.3.
- E. CPU 0.5 min/report; disk 0.25; network 0.33.

22. Which resource is the bottleneck, and what bound class does it represent?

- A. CPU; the service is memory-bound.
- B. Network; the service is compute-bound.
- C. All three; shared throughput makes them equally limiting.
- D. Disk; I/O-bound.
- E. CPU; processor work determines the system bottleneck.

23. What throughput ceiling does the current bottleneck set?

- A. 12 reports/hour from observed completions.
- B. 20 reports/hour from network demand.
- C. Exactly 15 reports/hour in this model.
- D. 48 reports/hour from disk busy time.
- E. 60 reports/hour from observation length.

24. A CPU upgrade halves CPU demand from 2 to 1 minute/report and changes nothing else. What happens to the modeled ceiling?

- A. It doubles to 30 reports/hour because the CPU became twice as fast.
- B. It rises to 60 reports/hour because CPU demand is now one minute.
- C. It falls to 12 reports/hour because the resource demands changed.
- D. It becomes 20 reports/hour because network is three minutes/report.
- E. It remains 15 reports/hour because disk demand is still 4 minutes/report.

25. A software change instead halves disk demand from 4 to 2 minutes/report. What becomes the bottleneck and ceiling?

- A. Network becomes the bottleneck; the ceiling becomes 20 reports/hour.
- B. Disk remains the bottleneck; the ceiling stays 15 reports/hour.
- C. CPU becomes the bottleneck; the ceiling becomes 30 reports/hour.
- D. There is no bottleneck; the ceiling becomes unlimited.
- E. All resources become bottlenecks; the ceiling becomes 12 reports/hour.

Scenario 4 continued — Recommend and remeasure

Continue with the improved report service: CPU demand is 2 minutes/report, disk demand is 2, network demand is 3, and the modeled ceiling is 20 reports/hour. The offered load is 10 reports/hour, with no growing backlog.

26. What sustained throughput should the service expect at this offered load?

- A. 20 reports/hour.
- B. 10 reports/hour.
- C. 30 reports/hour.
- D. 15 reports/hour.
- E. 2 reports/hour.

27. If offered load later rises to 19 reports/hour, why could latency become fragile even though 19 is below the 20/hour ceiling?

- A. Each report now requires 19 minutes of active service.
- B. The resource demands automatically double at 19 reports/hour.
- C. The system cannot complete a report below its ceiling.
- D. Near capacity, bursts can create slow-draining queues.
- E. Little's Law stops applying when utilization is below 100%.

28. Which recommendation is best supported by the original and revised demand calculations?

- A. Keep the disk change; then test the network.
- B. Buy the faster CPU because processor speed matters most.
- C. Replace all resources together, leaving no isolated comparison.
- D. Choose the most expensive change because price establishes performance.
- E. Skip observation because the modeled ceiling establishes the result.

29. Which remeasurement would most directly test whether the change helped the actual workload?

- A. Run a different workload and attribute the difference to disk.
- B. Use the vendor's maximum rate instead of running the workload.
- C. Repeat the same workload and boundary; compare rates, latency, and busy times.
- D. Measure one idle report and call it evidence of peak sustained throughput.
- E. Change the CPU, disk, and network together before collecting another log.

30. After the change, average whole-report latency falls. What does that observation establish by itself?

- A. The CPU alone caused the improvement.
- B. Disk demand fell by half during the test.
- C. Network demand cannot be the next bottleneck.
- D. Future workloads will improve by the same amount.
- E. Only the whole path improved; the internal cause remains unknown.